

# The Cumulative Distribution Function

2026-04-17

## Introduction

The cumulative distribution function (CDF) is the one representation that fits every random variable, whether discrete, continuous, or mixed. It encodes the distribution completely and supports every computation that a PMF or PDF does.

## Prerequisites

Random variables, basic probability.

## Theory

The CDF of a random variable  $X$  is

$$F_X(x) = P(X \leq x), \quad x \in \mathbb{R}.$$

Every CDF has the following properties:

1. **Monotonic non-decreasing:**  $x_1 \leq x_2 \Rightarrow F_X(x_1) \leq F_X(x_2)$ .
2. **Right-continuous:**  $\lim_{x \downarrow x_0} F_X(x) = F_X(x_0)$ .
3. **Limits:**  $\lim_{x \rightarrow -\infty} F_X(x) = 0$ ,  $\lim_{x \rightarrow +\infty} F_X(x) = 1$ .

Conversely, any function with these three properties defines a unique probability distribution on  $\mathbb{R}$ .

Key computations:

- Interval probability:  $P(a < X \leq b) = F_X(b) - F_X(a)$ .
- PDF (continuous  $X$ ):  $f_X(x) = F'_X(x)$ .
- PMF (discrete  $X$ ):  $p_X(x) = F_X(x) - F_X(x^-)$ , the jump size.
- Quantile function:  $Q_X(u) = \inf\{x : F_X(x) \geq u\}$ , the (generalised) inverse of  $F_X$ .

In R, each distribution family has a `p*()` function returning the CDF; the quantile function is `q*()`. The empirical CDF is obtained with `ecdf()`.

## Assumptions

None beyond a well-defined probability measure.

## R Implementation

```
library(ggplot2)

set.seed(2026)
x <- rnorm(200, mean = 75, sd = 10)

# Empirical CDF compared to theoretical CDF
Fn <- ecdf(x)
```

```

grid <- seq(50, 100, length.out = 401)
df <- data.frame(
  x      = grid,
  F_empir = Fn(grid),
  F_theor = pnorm(grid, mean = 75, sd = 10)
)

ggplot(df, aes(x = x)) +
  geom_step(aes(y = F_empir), colour = "#2A9D8F") +
  geom_line(aes(y = F_theor), colour = "#F4A261", linewidth = 1) +
  labs(y = "CDF", title = "Empirical (step) vs theoretical (line) CDF") +
  theme_minimal()

# Interval probability via the CDF
P_70_to_80 <- pnorm(80, 75, 10) - pnorm(70, 75, 10)
P_70_to_80

# Quantile: the value below which a given probability lies
qnorm(0.975, 75, 10)

```

## Output & Results

```

[1] 0.3829      # P(70 < X <= 80)
[1] 94.6        # 97.5th percentile

```

The empirical step-function is close to the theoretical normal CDF, and the differences narrow as  $n$  grows (Glivenko-Cantelli).

## Interpretation

In a paper, the CDF appears most often as a figure showing the empirical CDF against a theoretical reference (Q-Q plot is a rotation of this idea). Interval probabilities computed from the CDF are the standard form of reporting “percentage within the normal range”.

## Practical Tips

- The CDF is the unique object that characterises the distribution; two variables with the same CDF have the same distribution.
- Right-continuity vs left-continuity matters only at discrete atoms; the convention is right-continuous in most modern texts.
- The quantile function inverts the CDF; for continuous  $X$ , it is the ordinary functional inverse.
- The inverse-CDF method generates random draws:  $X = F^{-1}(U)$  where  $U \sim \text{Uniform}(0, 1)$ .
- For mixed distributions, the CDF is the most transparent representation; PDFs and PMFs require generalised definitions.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

### See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots